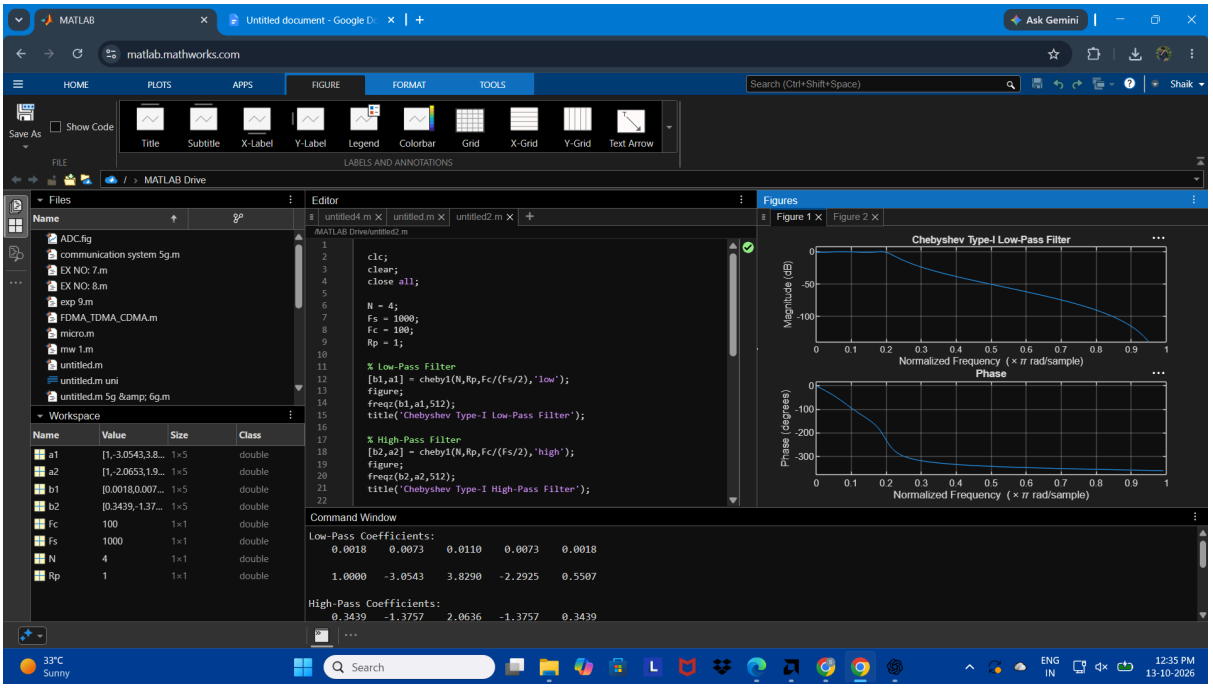
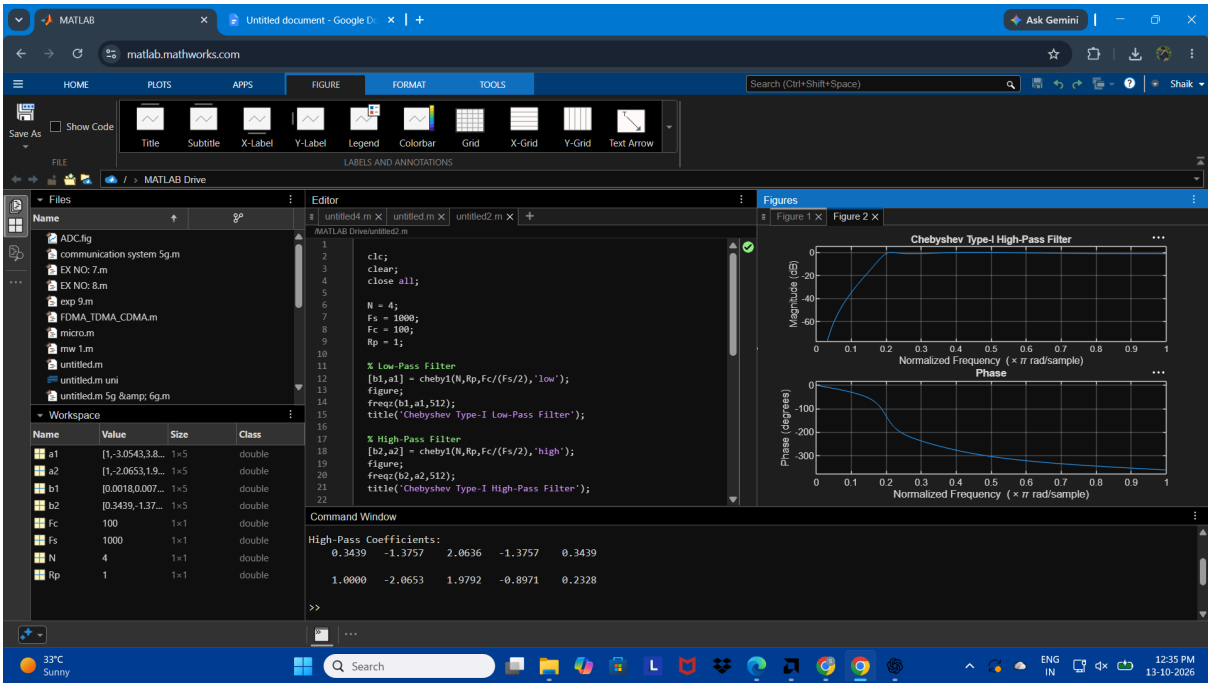
EXP NO:- 2

PROGRAM ➖

```
clc;
clear;
close all;
N = 4;
Fs = 1000;
Fc = 100;
Rp = 1;
% Low-Pass Filter
[b1,a1] = cheby1(N,Rp,Fc/(Fs/2),'low');
figure;
freqz(b1,a1,512);
title('Chebyshev Type-I Low-Pass Filter');
% High-Pass Filter
[b2,a2] = cheby1(N,Rp,Fc/(Fs/2),'high');
figure;
freqz(b2,a2,512);
title('Chebyshev Type-I High-Pass Filter');
disp('Low-Pass Coefficients:');
disp(b1);
disp(a1);
disp('High-Pass Coefficients:');
disp(b2);
disp(a2);
```

OUTPUT:-


MATLAB
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Y-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
Text Arrow
FILE
LABELS AND ANNOTATIONS
MATLAB Drive
Files
Name
ADC.fig
communication system 5g.m
EX NO: 7.m
EX NO: 8.m
exp 9.m
FDMA_TDMA_CDMA.m
micro.m
mw 1.m
untitled.m
untitled.m uni
untitled.m 5g & 6g.m
Workspace
Name
Value
Size
Class
a1
[1,-3.0543,3.8...
1×5
double
a2
[1,-2.0653,1.9...
1×5
double
b1
[0.0018,0.007...
1×5
double
b2
[0.3439,-1.37...
1×5
double
Fc
100
1×1
double
Fs
1000
1×1
double
N
4
1×1
double
Rp
1
1×1
double
Editor
untitled4.m
untitled.m
untitled2.m
/MATLAB Drive/untitled2.m
clc;
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title('Chebyshev Type-I High-Pass Filter');
Figures
Figure 1
Figure 2
Chebyshev Type-I Low-Pass Filter
Magnitude (dB)
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
Normalized Frequency (× π rad/sample)
Command Window
Low-Pass Coefficients:
0.0018 0.0073 0.0110 0.0073 0.0018
1.0000 -3.0543 3.8290 -2.2925 0.5507
High-Pass Coefficients:
0.3439 -1.3757 2.0636 -1.3757 0.3439
33°C
Sunny
Search
ENG
IN
12:35 PM
13-10-2026


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a2
[1,-2.0653,1.9...
1×5
double
b1
[0.0018,0.007...
1×5
double
b2
[0.3439,-1.37...
1×5
double
Fc
100
1×1
double
Fs
1000
1×1
double
N
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1×1
double
Rp
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1×1
double
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untitled2.m
/MATLAB Drive/untitled2.m
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Chebyshev Type-I High-Pass Filter
Magnitude (dB)
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
Normalized Frequency (× π rad/sample)
Command Window
High-Pass Coefficients:
0.3439 -1.3757 2.0636 -1.3757 0.3439
1.0000 -2.0653 1.9792 -0.8971 0.2328
>>
33°C
Sunny
Search
ENG
IN
12:35 PM
13-10-2026